Recurrent and Attention Models: A Two-Column Study

Transformer architectures have recently replaced recurrent networks in many sequence modelling tasks. The transformer relies entirely on self-attention, which computes interactions between all pairs of positions in a sequence and therefore captures long-range dependencies without recurrence.

Nevertheless, recurrent models remain attractive for streaming applications. A Long Short-Term Memory network processes observations incrementally, maintaining a compact representation of the past, and its computational requirements grow linearly with sequence length rather than quadratically.

Our experiments compare both families on trajectory prediction for commercial aircraft. We report displacement errors at several horizons, discuss computational requirements, and analyse failure cases where the transformer hallucinates manoeuvres that never occur in the surveillance recordings.

Interestingly, hybrid architectures that combine convolutional feature extraction with recurrent aggregation achieve competitive performance while requiring considerably fewer parameters. These observations suggest that architectural inductive biases still matter in data-limited regimes.

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